

Graphical relationships and Inverse Functions — Problem Set

Warm-up

1. Find the equation of the inverse function $f^{-1}(x)$ for the linear function $f(x) = 3x - 5$.
2. State the natural domain and range of the inverse sine function, $y = \sin^{-1} x$.
3. State the natural domain and range of the inverse cosine function, $y = \cos^{-1} x$.
4. State the natural domain and range of the inverse tangent function, $y = \tan^{-1} x$.
5. If $f(x) = \frac{1}{x+2}$, write down the equation of the reciprocal function $y = \frac{1}{f(x)}$.
6. Given the point $(3, -1)$ lies on the graph of a one-to-one function $y = f(x)$, state the coordinates of the corresponding point on the graph of $y = f^{-1}(x)$.
7. Sketch the graph of $y = |x^2 - 4|$ by transforming the graph of $y = x^2 - 4$.
8. Sketch the graph of $y = \ln(|x|)$ by transforming the graph of $y = \ln(x)$.
9. Find the domain of the function $f(x) = \sqrt{x - 3}$.

Skill-building

1. Consider the function $f(x) = \frac{a-x}{a+x}$ for $x \neq -a$. Show that $f(f(x)) = x$ for all x in the domain, and explicitly state what this implies about the relationship between $f(x)$ and $f^{-1}(x)$.
2. Sketch the graph of $y = \frac{1}{f(|x|)}$ where $f(x) = (x - 2)(x - 4)$, clearly indicating all vertical asymptotes, horizontal asymptotes, and stationary points.
3. Let $f(x) = \ln(x + \sqrt{x^2 + 1})$ for all $x \in \mathbb{R}$.
 1. Find an explicit expression for the inverse function $y = f^{-1}(x)$.
4. The function $f(x) = e^{2x} - 4e^x + 3$ is restricted to a domain $x \leq k$ such that it possesses a unique inverse function $f^{-1}(x)$. Find the maximum possible value of k .
5. Sketch the graph of $y = \tan^{-1}(\tan x)$ over the domain $-2\pi \leq x \leq 2\pi$, carefully indicating the positions of all discontinuities and gradients.
6. For the function $f(x) = |x - 1| - |x + 2|$:
 1. Sketch the graph of $y = f(x)$.
 2. Sketch the graph of $y = \frac{1}{f(x)}$, showing all features.
7. Let $f(x) = \frac{x}{\sqrt{1-x^2}}$ for $-1 < x < 1$. Find the algebraic equation for $f^{-1}(x)$ and state its domain and range.

8. A function is defined piecewise by:

$$f(x) = \begin{cases} -x^2 - 2x & \text{for } x < -1 \\ |x| & \text{for } x \geq -1 \end{cases}$$

1. Sketch $y = f(x)$.
2. Sketch $y = f(x) + f(-x)$, showing all intercepts and sharp edges.

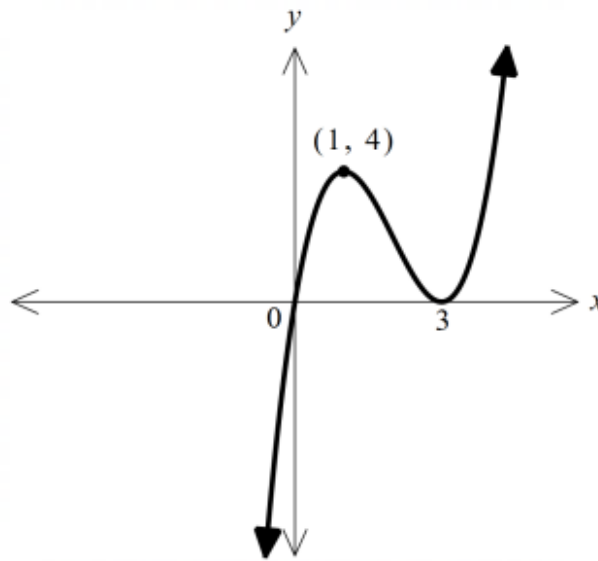
Easier Exam Questions

1. (Baulkham Hills 2020 Q12b)

The diagram shows the graph of a function $y = f(x)$.

- (i) Draw the graph of $y = \frac{1}{f(x)}$ showing all important features.

3



- (ii) Draw the graph of $y = f(|x|)$ showing all important features.

2

2. (Girraween (2022) Q6c)

The diagram shows the graph of the function $f(x) = (x + 2)(1 - x)$.

Sketch the following:

(i) $y = \frac{1}{f(x)}$

3

(ii) $y = |f(x)|$

2

(iii) $y = f(|x|)$

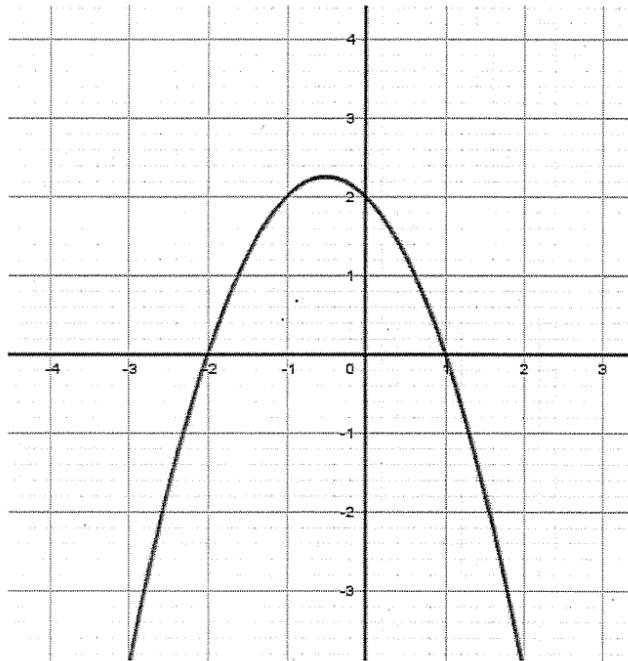
2

3. (James Ruse (2022) Q9)

For which of the following conditions does the graph of

$$y = \frac{1}{ax^2 + bx + a} \text{ for } a \in \mathbb{R} \setminus \{0\}$$

have exactly two asymptotes?



- (A) $-2 < \frac{a}{b} < 2$
- (B) $-2 \leq \frac{a}{b} \leq 2$
- (C) $-1 < \frac{b}{a} < 1$
- (D) $\frac{a}{b} = \frac{1}{2}$

4. (Baulkham Hills (2019) Q8)

Which of the following is the range of the function $f(x) = \sin^{-1} x + \tan^{-1} x$?

- (A) $-\pi < y < \pi$
- (B) $-\pi \leq y \leq \pi$
- (C) $-\frac{3\pi}{4} \leq y \leq \frac{3\pi}{4}$
- (D) $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$

5. (Baulkham Hills (2023) Q2)

Which of the following is the natural domain of $f(x) = \cos^{-1}\left(\frac{x-a}{b}\right)$, where $b > 0$?

- (A) $[-1, 1]$
- (B) $[a - b, a + b]$
- (C) $[a - 1, a + 1]$
- (D) $[-b, x + b]$

6. (Caringbah (2019) Q13b)

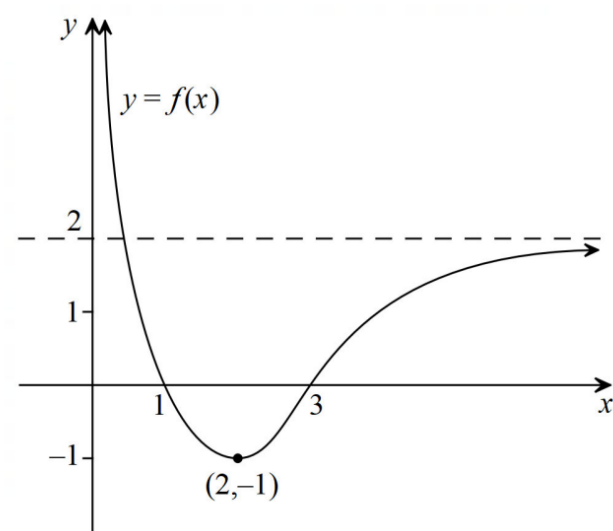
Find the equation of the inverse $f^{-1}(x)$ for $f(x) = \frac{1-x}{3+x}$.

2

Harder Exam Questions

1. (Baulkham Hills (2022) Q12a)

Consider the graph of $y = f(x)$ below.



On separate axes, sketch each of the following, showing all important features.

3

$$y = \frac{1}{f(|x|)}$$

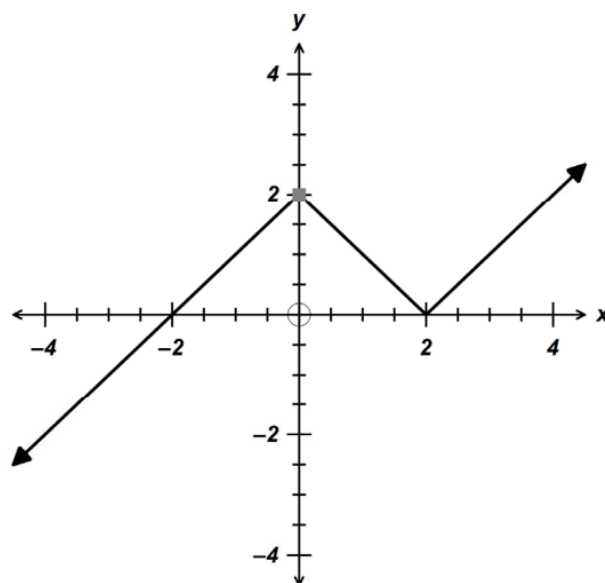
2. (Blacktown Boys (2023) Q13(e))

Sketch the graph of $f(x) = |x|^4 + 4|x| - 3$, showing all key features.

2

3. (Presbyterian (2022) Q12d)

The graph below shows $y = f(x)$ where $f(x) = x + 2$ for $x < 0$ and $f(x) = |x - 2|$ for $x \geq 0$

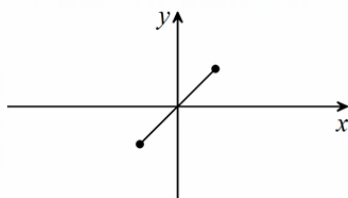


Sketch the graph of $y = f(x) + f(-x) + 1$, carefully showing any intercepts on the co-ordinate axes. 3

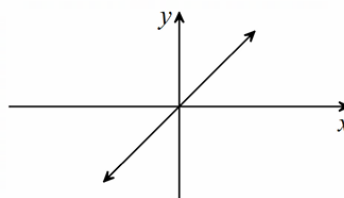
4. (Baulkham Hills (2022) Q2)

Which of the following is the best representation of the graph of $y = \arcsin(\sin x)$?

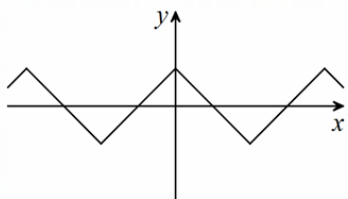
A.



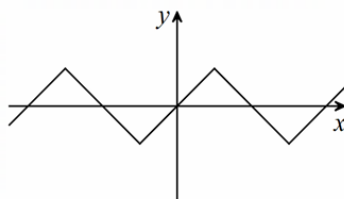
B.



C.

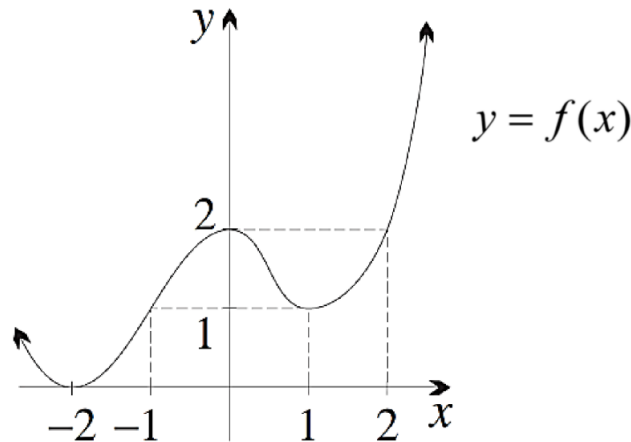


D.



5. (Fort St (2022) Q4)

Given the graph of the quartic $y = f(x)$ below, which of the following restrictions on the domain would allow the function $f(x)$ to have an inverse function $y = f^{-1}(x)$?



- (A) $[-2, 1]$
- (B) $[-2, 2]$
- (C) $[0, \infty)$
- (D) $[-2, 0]$

6. (Fort St (2021) Q4a)

(a) If $f(x) = \ln(ex)$ and $g(x) = e^{x-1}$ show that:

(i) $f(g(x)) = x$ 2

(ii) $g(f(x)) = x$ 2

(iii) What do the results of (i) and (ii) indicate about $f(x)$ and $g(x)$? 1

7. (Knox (2019) Q14b)

What is the domain of the function $g(x) = \cos^{-1}(\ln(x))$? 2

8. (Knox (2019) Q14d)

Let $f(x) = e^x - \frac{1}{e^x}$ for all real values of x .

(ii) Sketch the graph of $y = f(x)$. 2

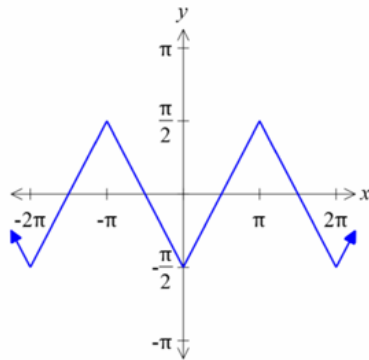
(iii) On a separate diagram, sketch the graph of the inverse function. 1

(iv) Find an expression for $y = f^{-1}(x)$ in terms of x . 3

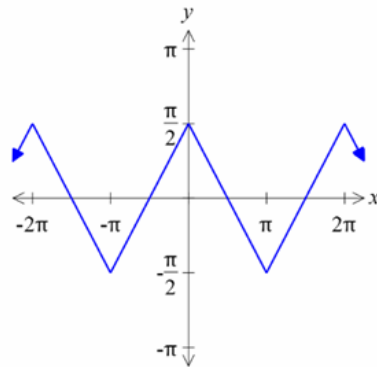
9. (Knox (2022) Q9)

Which graph best represents the function $\sin^{-1}(-\cos(x))$?

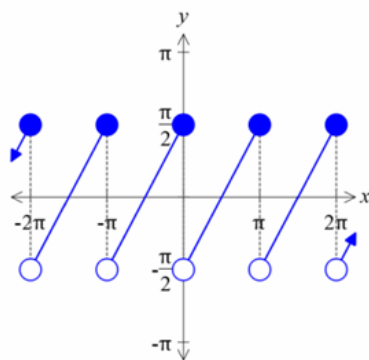
A.



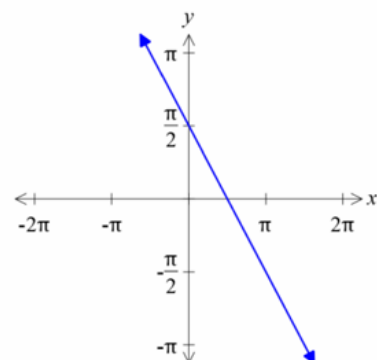
B.



C.



D.



10. (Presbyterian (2023) Q1)

The function, $f(x)$, is one-to-one and has domain $[0, 5]$ and range $(-2, 5)$. Which of the following having correctly describes the domain and range of its inverse function, $f^{-1}(x)$?

	Domain	Range
(A)	$(-2, 5]$	$[0, 5)$
(B)	$[-2, 5)$	$(0, 5]$
(C)	$[0, 5)$	$(-2, 5]$
(D)	$(0, 5]$	$[-2, 5)$

11. (Sydney Boys (2019) Q15d)

The function $f(x)$ is defined by $f(x) = \frac{1-x}{1+x}, x \neq -1$. Given that $f^2(x) = f(f(x))$, by first finding the simplest expression for $f^2(x)$, show that $f^{2020}(x) = x, x \neq -1$. **2**

12. (Sydney Girls (2022) Q4a(i))

(a) Consider the function $f(x) = \frac{1}{2}(x-3)^2 + 4$.

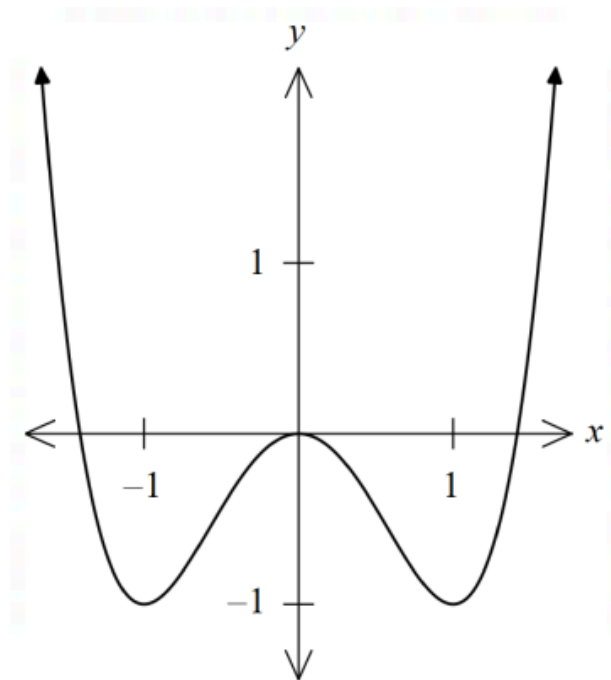
(i) Sketch the parabola $y = f(x)$, showing clearly the coordinates of the vertex. Hence write down the largest domain containing the value $x = 2022$, for which the function has an inverse function $f^{-1}(x)$. **2**

(ii) For the domain found in part (i), find $f^{-1}(x)$. **2**

(iii) Let a be a real number NOT IN the domain found in part (i). By considering the symmetry of the parabola, or otherwise, find $f^{-1}(f(a))$ **2**

13. (James Ruse (2024) Q5)

The function $f(x) = -\sqrt{1 - \sqrt{1 + x}}$ has an inverse $f^{-1}(x)$. The graph of $y = f^{-1}(x)$ forms part of the curve $y = x^4 - 2x^2$.



How many points do the graphs of $y = f(x)$ and $y = f^{-1}(x)$ have in common?

- (A) 1
- (B) 2
- (C) 3
- (D) 4

14. (North Sydney Girls (2019) Q9b)

Consider the function $f(x) = 2^x - 2^{-x}$.

(i) Sketch $y = 2^x$ and $y = -2^{-x}$ on the same set of axes, and use the method of addition of ordinates to sketch the graph of $y = f(x)$. **2**

(ii) Explain, with reference to the graphs of $y = 2^x$ and $y = -2^{-x}$, why the function $f(x)$ has an inverse function. **1**

(iii) Replacing $f(x)$ by y , show that this equation can be written as: $2^{2x} - y \cdot 2^{-x} - 1 = 0$. **1**

(iv) Hence find the equation of the inverse function $y = f^{-1}(x)$. **3**